

FINAL PROJECT REPORT

Delhi Air Quality Analysis (2021 - 2024)

Understanding Delhi's Pollution Problem

1. Objective

To analyze Delhi's daily air pollution data and find out:

- Which season has the worst air quality?
 - How much does Diwali affect pollution?
 - What are the main reasons for bad air?
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2. Tools & Technologies Used

- Excel → For initial data checking
 - Python (Pandas) → Data analysis
 - Plotly → Interactive charts
 - Jupyter Notebook
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3. About the Data

- Daily Air Quality Data of Delhi
 - Time Period: 2021 to 2024
 - Columns: Date, AQI, PM2.5, PM10, NO2, SO2, CO, Ozone etc.
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4. Key Insights (Most Important Part)

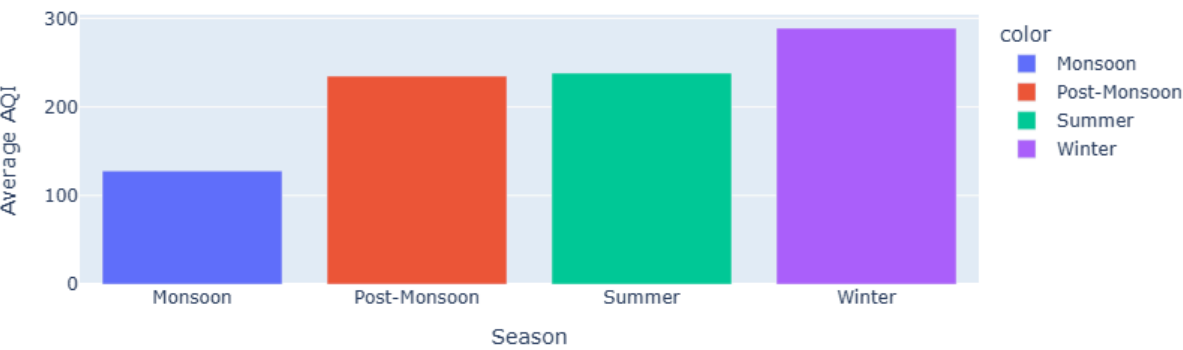
1. **Winter Season is the Most Dangerous** Pollution level is highest in Winter (November to February).
2. **Diwali Makes Air Much Worse** Average AQI is significantly higher in October and November.
3. **PM2.5 and PM10 are Main Culprits** These two pollutants contribute the most to bad AQI.
4. **Many Severe Days** AQI crossed 400 (Severe category) multiple times, especially in winter.
5. **Monsoon Brings Relief** Rain helps in reducing pollution levels.

5. Visualizations

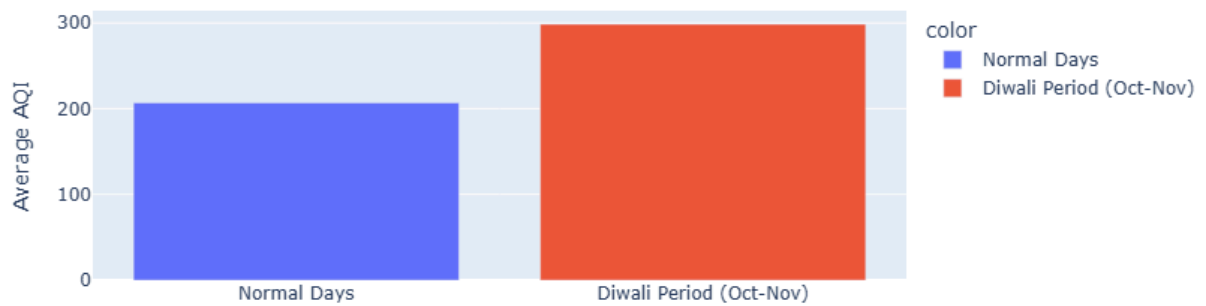
Delhi Air Quality Trend (2021-2024)



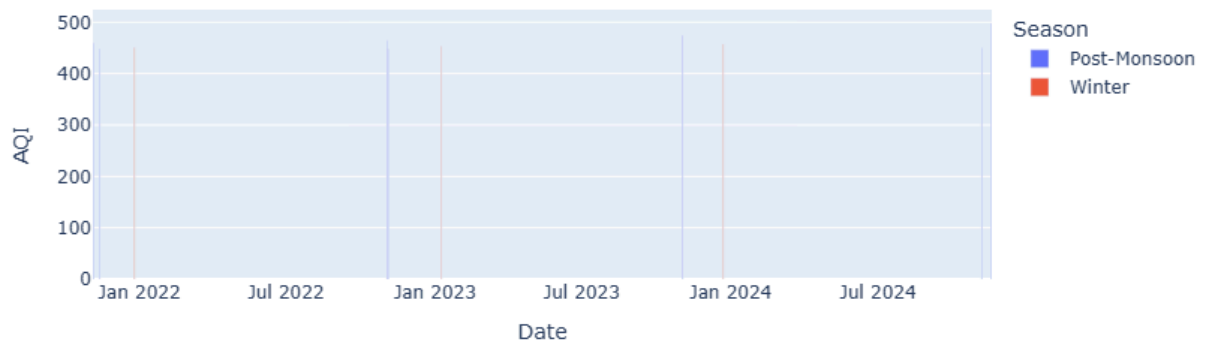
Delhi Pollution in Different Seasons



How Much Worse is Air During Diwali?



Top 10 Worst Air Quality Days in Delhi



Yearly AQI Trend Chart

- Season-wise AQI Bar Chart
- Diwali Impact Chart
- Top 10 Worst Days Chart

Tip: Keep 4–5 best charts only.

6. Conclusion

Air pollution in Delhi is a serious health issue. It becomes extremely dangerous during winter and festival seasons. The main causes are stubble burning, fireworks, traffic, and weather conditions.

This project shows that data analysis can help us understand real-world problems and support better decision making.

7. What I Learned

- How to load and clean data using Python
- Creating charts and visualizations
- Finding meaningful insights from raw data
- Importance of time series analysis

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